Misplaced Blame in the Corn Belt? Testing the Causal Impact of Hyperscale Data Center Expansion on Public Drinking Water Nitrate Concentrations in Iowa

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Abstract

During the summers of 2025 and 2026, historic nitrate spikes in the Raccoon and Des Moines rivers forced Central Iowa Water Works (CIWW) to impose the first mandatory outdoor lawn-watering restrictions in Des Moines metropolitan history for over 600,000 residents. Because these emergency restrictions coincided with an unprecedented wave of hyperscale data center construction across Polk, Dallas, and Linn counties—and because homeowners faced water shutoffs for irrigating lawns while evaporative cooling towers at Microsoft, Meta, Apple, and Google continued operating—viral social media commentary, local news coverage, and public testimony at county zoning hearings rapidly linked data centers to Iowa's drinking water nitrate crisis. By late 2026, twelve Iowa counties and multiple municipalities had enacted or advanced data center moratoriums, frequently citing water quality alongside water quantity concerns. If data center construction or operation elevated nitrate concentrations (mg/L as N) in Public Water Supplies (PWS), that link would provide urgent public-health justification for halting data center expansion. Conversely, if data centers have zero causal effect on drinking water nitrate, attributing river nitrate crises to digital infrastructure diverts scarce regulatory capital and civic attention away from the actual drivers of nitrate contamination.

We execute the first comprehensive econometric test of the Null Hypothesis ($H_0 : \beta = 0$) across all 70 tracked data center facilities in Iowa (6.2 GW of reported power across 18 counties) linked to a balanced 22-year quarterly panel ($N = 4,176$ observations across 48 Community Public Water Supplies from 2005Q1–2026Q3). Using Two-Way Fixed Effects (TWFE), Callaway and Sant'Anna (2021) staggered difference-in-differences, and Sun and Abraham (2021) stacked event studies across three distinct treatment timing definitions (T_{permit}, $T_{\text{construct}}$, and $T_{\text{operational}}$) and three spatial scales, we find zero effect of data center development on PWS nitrate concentrations. In our primary 10 km buffer specification clustered by HUC-10 watershed, the construction grading effect is $\hat{\beta}_{\text{construct}} = +0.0013$ mg/L as N (SE $= 0.0178, p = 0.9430$) and the operational cooling activation effect is $\hat{\beta}_{\text{operational}} = -0.0124$ mg/L as N (SE $= 0.0142, p = 0.3899$). Two One-Sided Tests (TOST) formally confirm statistical equivalence to zero ($p_{\text{TOST}} < 0.0001$) within both $\Delta = \pm 0.25$ mg/L (2.5% of the EPA Maximum Contaminant Level) and $\Delta = \pm 0.05$ SD, with 90% confidence intervals confined inside $[-0.037, +0.032]$ mg/L. Stratifying by aquifer geology and precipitation anomalies demonstrates that post-drought spring flushing of agricultural tile drainage ($+0.1778$ mg/L per $+1$ SD precipitation anomaly, $p < 0.01$)—not industrial cooling demand—drives Iowa's municipal nitrate crises.

Keywords: Data Centers; Drinking Water Nitrate; Staggered Difference-in-Differences; Equivalence Testing (TOST); Jordan Aquifer; Agricultural Nonpoint Runoff; Environmental Policy Moratoriums.
JEL Classification: Q25, Q53, Q58, R14, L86, C23.

1 Introduction: Media Outrage, Lawn-Watering Bans, and the Moratorium Wave

On June 12, 2025, Central Iowa Water Works (CIWW)—the regional drinking water production authority serving more than 600,000 residents across Des Moines, West Des Moines, Waukee, Ankeny, Urbandale, Clive, Grimes, Johnston, Norwalk, and Polk City—issued the first mandatory Stage III outdoor lawn-watering ban in the metropolitan area's history [5, 22]. Raw water nitrate-nitrogen (NO_3-N) concentrations had surged to 14.39 mg/L in the Raccoon River and 14.04 mg/L in the Des Moines River on the day the ban was declared, eventually peaking at 20.55 mg/L in the Raccoon River and 17.15 mg/L in the Des Moines River during summer 2025 [12, 36]. Even with CIWW's seven-vessel ion-exchange Nitrate Removal Facility at the Fleur Drive Treatment Plant running at maximum capacity at an operating cost of $\$16,000$ per day, finished tap water distributed to homes reached 9.17 mg/L from Fleur Drive and 9.04 mg/L from the L.D. McMullen plant—less than 1 mg/L below the U.S. Environmental Protection Agency (EPA) Maximum Contaminant Level (MCL) of 10.0 mg/L as N established to prevent infant methemoglobinemia ("blue baby syndrome") [21, 36, 17].

Barely ten months later, in May and June 2026, the crisis returned with even greater persistence. By June 8, 2026, nitrate concentrations in the Raccoon River had exceeded the 10.0 mg/L federal threshold for 136 days, the Des Moines River infiltration gallery had remained above 10.0 mg/L for nearly 90 consecutive days (an all-time utility record), and CIWW was forced to draw 3.5 million gallons per day (MGD) from emergency winter reserves in the Dale Maffitt Reservoir before running the Nitrate Removal Facility for 193 consecutive days through July 27, 2026 [6, 23, 7]. During a June 2026 press briefing, Des Moines Water Works CEO Amy Kahler warned that customer demand was projected to hit 98% of reduced nitrate-treatment capacity and announced that households continuing to water lawns two days after a warning tag would have their municipal water service physically shut off [23].

Within hours of both the June 2025 mandatory ban and the May 27, 2026 Stage II conservation alert, public frustration across local news comment sections, neighborhood forums, and Des Moines social media threads converged on a highly visible suspect: Iowa's booming hyperscale data center industry [1, 25, 38]. Residents driving past massive data center campuses along Interstate 80 in Altoona (Meta Platforms), Interstate 35 and Veterans Parkway in West Des Moines (Microsoft), Grand Prairie Parkway in Waukee (Apple), and Highway 30 in Cedar Rapids (Google and QTS) questioned why ordinary citizens were threatened with water disconnection to prevent a nitrate MCL violation while billion-dollar tech campuses continued evaporating millions of gallons of municipal water to cool artificial intelligence servers [1, 40, 46].

The anecdotal conflation of data centers, summer water restrictions, and drinking water nitrate levels grew so pervasive that municipal utilities and county emergency managers were forced to issue extraordinary public rebuttals:

- On June 12, 2025, Polk County Emergency Management published an official Nitrate Emergency FAQ explicitly addressing the viral question: *"Q: Are data centers being asked to reduce water usage? A: No. Records show that data centers use 2–6% of the system capacity versus outdoor water usage which composes 40% or more of our system capacity during peak watering season"* [35].
- On June 17, 2025, Axios Des Moines published a dedicated fact-check titled *"Data centers are not to blame for central Iowa water crisis,"* after residents repeatedly blamed Microsoft's West Des Moines facilities for the nitrate-triggered lawn ban [1].
- On June 1, 2026, following renewed public outcry over the May 27, 2026 Stage II nitrate alert, [West Des Moines Water Works] released a formal white paper (*"Just the Facts: Data Centers & Water"*) stating verbatim: *"Data centers do not affect the quality of the source water treated at the A.C. Ward Municipal Water Treatment Plant or any other facility owned by CIWW, nor did the water*

provided to data centers lead to the metro's first-ever lawn-watering ban in 2025... In July 2025, the watering ban was in effect for 18 days. Data center use in July comprised only 0.9% of the water produced by CIWW" [46].

- Simultaneously, the City of Waukee updated its municipal conservation portal with a dedicated section (*"Are data centers using a lot of water?"*) clarifying that data centers account for less than 1% of Waukee's operational water supply [45].

Yet utility assurances have not settled the policy debate. Instead, anecdotal fears linking data centers to water degradation and depletion have fueled a sweeping regulatory backlash across Iowa. Between February and October 2026, at least **twelve Iowa counties**—Linn County (which enacted an 18-month moratorium on EU-3 large-scale data center rezonings on July 1, 2026 through January 1, 2028), Dubuque County (which adopted a 12-month moratorium under Resolution 26-151 on July 6, 2026), Adair County (Resolution 2026-26 on March 18, 2026), Franklin County, Audubon County, Dickinson County, Des Moines County, Lee County, Plymouth County, Sioux County, Woodbury County (Resolution 14064 on June 23, 2026), and Ida County—alongside municipalities such as Peosta (six-month moratorium enacted June 23, 2026 and extended September 22, 2026 through June 2027) and Marshalltown enacted or advanced formal pauses on data center development [39, 30, 13, 34]. When the Linn County Board of Supervisors approved a Memorandum of Understanding with Google on October 28, 2026 for an independent water study during its moratorium, Supervisor Kirsten Running-Marquardt emphasized the public's dual concern: *"Residents continue to tell us that water quality, water availability, and planning for floods and droughts are priorities"* [31].

The Policy Stakes and the Opportunity Cost of Misdirected Attention: This controversy presents a high-stakes empirical question with asymmetric policy consequences. On one hand, if the construction, land grading, or cooling-water operations of data centers causally elevate nitrate concentrations in nearby Public Water Supplies, then citizen calls for data center moratoriums carry urgent public-health weight. Epidemiological research conducted in Iowa by the University of Iowa College of Public Health demonstrates that chronic exposure to drinking water nitrate even at half the federal limit (> 5.0 mg/L as N) is consistently associated with elevated risks of colorectal, thyroid, ovarian, and kidney cancers, as well as neural tube and limb-deficiency birth defects [43].
On the other hand, **if data centers have zero causal impact on drinking water nitrate concentrations ($H_0 : \beta = 0$), blaming data centers inflicts a severe environmental and public-policy opportunity cost**. Every hour of county supervisor hearings, municipal legal drafting, and environmental advocacy spent attempting to curb nitrate pollution by pausing data centers diverts scarce political capital, regulatory bandwidth, and research funding away from the actual, overwhelming source of Iowa's nitrate crisis: nonpoint agricultural nitrogen loading from 23.9 million acres of fertilized corn and soybean cropland accelerated by subsurface tile drainage and climate-driven drought-to-deluge river flushing [27, 22, 20].

In this paper, we subject the popular media and anecdotal hypothesis to rigorous econometric falsification and statistical equivalence testing. Rather than relying on simple before-and-after comparisons or single-utility case studies, we assemble a complete longitudinal inventory of all **70 tracked data center facilities and projects in Iowa** (6.2 GW of reported power across 18 counties per US Data Center Map) and link them spatially and hydrologically to a balanced 22-year quarterly panel ($N = 4,176$ observations across 48 verified Iowa Community Public Water Supplies from 2005Q1 through 2026Q3 per Iowa DNR Source Water Protection Tracker). We test whether data center permitting (T_{permit}), construction land disturbance ($T_{\text{construct}}$), or operational water withdrawal ($T_{\text{operational}}$) alters PWS nitrate concentrations across three spatial scales (same municipality, 5/10/20 km radial buffers, and shared/upstream HUC-10 watersheds).

Because failing to reject a traditional null hypothesis ($p > 0.05$) cannot distinguish between a true zero effect and an imprecise estimate, we pair modern staggered Difference-in-Differences estimators—Callaway and Sant'Anna (2021) and Sun and Abraham (2021) stacked event studies—with formal **Two One-Sided Tests (TOST) of equivalence** and ex-post Minimum Detectable Effect ($\text{MDE}_{80\%}$) calculations. Across every specification, estimator, and aquifer stratum, we reject non-equivalence at $p_{\text{TOST}} < 0.0001$, proving that the causal impact of data centers on Iowa drinking water nitrate is bounded tightly inside ± 0.04 mg/L as N—less than 0.4% of the federal MCL.

2 Institutional Background: Anatomy of a Hydrological Conflation

To understand why public discourse connected data centers to the 2025–2026 nitrate restrictions, one must distinguish between **source-water contaminant loading** (how nitrate enters rivers and aquifers) and **treatment-plant hydraulic capacity** (how utilities remove nitrate during peak summer demand).

2.1 Why High River Nitrates Trigger Summer Lawn-Watering Bans

Under the Safe Drinking Water Act, Community Water Systems must deliver finished tap water with nitrate-nitrogen strictly at or below 10.0 mg/L as N [16, 18]. Standard municipal water treatment processes—coagulation, flocculation, lime softening, sedimentation, granular activated carbon filtration, and chlorination—do **not** remove dissolved nitrate ions (NO_3^-) [10, 29]. When raw source water in the Raccoon or Des Moines rivers exceeds 10.0 mg/L as N, Central Iowa Water Works must rely on three specialized mitigation mechanisms [5, 23]:

1. **Ion-Exchange Nitrate Removal (Fleur Drive Treatment Plant):** Passing river/gallery water through seven specialized anion-exchange resin vessels that swap nitrate ions for chloride ions ($R\text{-Cl} + NO_3^- \rightarrow R\text{-}NO_3 + Cl^-$).
2. **Reverse Osmosis and Ultrafiltration (Saylorville and Grimes Plants):** Forcing water through semi-permeable membranes that reject nitrate but reject 15% to 25% of feedwater volume as brine concentrate [29, 23].
3. **Low-Nitrate Source Blending:** Diluting high-nitrate river water with low-nitrate water from deep confined bedrock wells (such as the Cambrian-Ordovician Jordan Aquifer at West Des Moines' A.C. Ward plant, which tests at 0.40 mg/L as N, or the Silurian Aquifer in Iowa City) or off-channel reservoirs (Dale Maffitt Reservoir) [36, 41, 23].

Crucially, as [West Des Moines Water Works (2026)] documented, routing water through ion-exchange vessels and reverse-osmosis membranes **reduces the net hydraulic throughput capacity of the Fleur Drive and McMullen plants by 25% or more** [46]. During cool spring months when residential water demand is low, a 25% reduction in maximum plant throughput leaves ample treated water for the metro. However, when a June or July heatwave ($> 80°$F) strikes while river nitrate remains at 14 to 20 mg/L, residential lawn irrigation surges to 30% to 40% of total metropolitan water demand (and up to 50% during dry Augusts), pushing total system demand past 90% to 98% of the nitrate-constrained plant capacity [46, 23].

Because data centers also activate evaporative cooling towers when ambient wet-bulb temperatures rise above 80°F, the public observed two simultaneous summer phenomena—(i) mandatory bans on residential lawn watering due to "nitrate treatment capacity" and (ii) visible steam plumes rising from data center cooling towers—and inferred a causal link [25, 39]. Yet in physical reality, Microsoft's five operational West Des Moines campuses accounted for just 0.3% to 0.9% of total CIWW water production during the July 2025 ban, Apple accounted for $< 1\%$ of Waukee's supply, and Meta's Altoona campus drew 0% from CIWW because Altoona operates its own deep Jordan Aquifer wells [46, 25, 45].

2.2 The Scale and Geography of Iowa's Data Center Buildout

While data centers represent a small fraction of metropolitan surface-water demand, their rapid capital and electrical expansion across Iowa makes rigorous environmental evaluation essential. According to the Common Sense Institute's October 2026 study and US Data Center Map's October 2026 directory, Iowa hosts **70 tracked data center facilities and projects** with 2,324 MW of capacity currently operating (ranking second in the Midwest behind Ohio and first per capita at 71.8 MW per 100,000 residents) and 5,257 MW in the development pipeline [9, 44]. Between 2011 and 2026, operators invested $15.1 billion in Iowa data center construction and equipment, with another $38.0 billion announced for 2027–2031 [9].

Table 1 summarizes our longitudinal facility dataset (`iowa_data_centers_panel.csv`), which reconciles facility records from US Data Center Map, Baxtel, Common Sense Institute, Server Country, Clean Energy Districts, and Iowa's AI Future.

3 Data Engineering, Regulatory Timelines, and Panel Construction

3.1 Establishing the Three Lifecycle Treatment Dates ($T_{\text{permit}}, T_{\text{construct}}, T_{\text{operational}}$)

Static facility indicators cannot distinguish between potential soil-disturbance effects during initial construction and water-withdrawal effects during facility operation. For each of the 70 facilities in `iowa_data_centers_panel.csv`, we cross-referenced Iowa Economic Development Authority (IEDA) High Quality Jobs and sales-tax exemption board awards (Iowa Code §423.3(95)), municipal building permit records in Altoona, West Des Moines, Waukee, Council Bluffs, and Cedar Rapids, Iowa DNR NPDES General Permit No. 2 construction stormwater filings (required for land disturbance > 1 acre), and Iowa DNR Water Use Permits (required for withdrawals $> 25,000$ gallons per day) to establish three distinct treatment dates:

1. **Permit / Entitlement Date (T_{permit}):** The date of IEDA tax-incentive approval, municipal development agreement passage, or DNR stormwater Notice of Intent (NOI) filing (e.g., June 19, 2007 for Google Council Bluffs Campus 1; August 21, 2008 for Microsoft Project Mountain; April 19, 2013 for Meta Altoona Building 1; August 24, 2017 for Apple Waukee; April 28, 2024 for Google Cedar Rapids) [26, 11, 39].
2. **Construction / Grading Start Date ($T_{\text{construct}}$):** The date of physical site stripping, mass grading, and warm-shell foundation work (when topsoil disturbance could theoretically release mineralized soil nitrogen into surface runoff).
3. **Operational / Water Withdrawal Activation Date ($T_{\text{operational}}$):** The date of certificate of occupancy, server energization, and commencement of cooling-water withdrawal and NPDES blowdown discharge (e.g., May 2009 for Google CB1; June 2011 for Microsoft Mountain; November 2014 for Meta Altoona B1; November 2024 for Apple Waukee Phase 1) [26, 9].

3.2 Public Water Supply Nitrate Data, Stoichiometric Standardization, and Censoring

We retrieve longitudinal finished and source-water nitrate measurements for **48 Iowa Community Public Water Supplies (CWS)** spanning 2005Q1 through 2026Q3 (87 quarters$\times$48 PWSs $= 4,176$ PWS-quarter observations 22 years $\times$ 48 PWSs $= 1,056$ PWS-year observations). System identifiers (PWSIDs), populations served, and primary aquifer classifications are verified directly against the Iowa DNR Source Water Protection

Table 1: Longitudinal Inventory of Tracked Data Center Facilities in Iowa by County (As of Late 2026)

County	**Sites**	**Oper.**	**Pipeline**	**Power (MW)**	**Major Operators & Tracked Campuses**	**Primary Cooling Water & Receiving Hydrogeology**
Polk	17	12	5 (2/3/0)	2,580.3	Meta Altoona (B1–B12), Microsoft (Mountain, Alluvion, Ginger East), Tract, Cielo, LightEdge, Edged	Deep Jordan Aquifer (Altoona PWS) & Blended Jordan/Alluvial/Surface (WDMWW / CIWW)
Dallas	6	4	2 (0/2/0)	1,704.0	Apple Waukee, Microsoft (Ginger West, Van Meter Vision Park), COPT Defense, US Signal	Purchased Surface/Alluvial (Waukee/CIWW) & Raccoon River Alluvial (Van Meter PWS)
Linn	9	3	6 (3/3/0)	614.0†	Google Big Cedar & Palo, QTS Big Cedar P1–P3, CLOP, Enseva, ImOn, Aureon	Cedar River Collector Wells (Cedar Rapids PWS) & Closed-Loop Water-Free (QTS/CLOP)
Pottawattamie	5	3	2 (1/1/0)	509.7	Google Council Bluffs (Campus 1, Southlands 2E/2W), CyrusOne OCB1, Edged Lola	Missouri River Surface & Alluvial Wells (Council Bluffs Water Works)
Warren & Madison	4	2	2 (0/2/0)	263.0	Microsoft Osmium (Warren/Madison), Microsoft Ruthenium, Tract Norwalk	Blended Jordan/Alluvial/Surface (WDMWW / Norwalk PWS)
Scott	4	3	1 (0/1/0)	4.0‡	Bluebird ColoHub, US Signal, South Front, Meta Davenport (114 MW)	Mississippi River Surface Water (Iowa-American Davenport)
Other 11 Counties	25	13	12 (0/9/3)	525.5	US Secure Hosting (Jones), LightEdge (Story), Applied Digital (Adair), QTS (Clinton), InfoBunker (Boone), PathOne (Sioux/Ida)	Silurian Bedrock (Monticello), Skunk Alluvial (Ames), Dakota/Alluvial (Adair, Hull, Galva, Boone)
Statewide Total	**70**	**40**	**30 (6/21/3)**	**6,200.5**	**All 70 Tracked Iowa Facilities**	**Linked to 48 Community PWSs (32 HUC-10s)**

Notes: Facility counts and deduplicated published power figures align with US Data Center Map (Oct 2026) (40 operational, 6 under construction, 21 planned, 3 paused; 6.2 GW reported power). †In Linn County, third-party Cleanview/CSI pipeline estimates include an additional 616 MW for QTS Big Cedar and 615 MW for Google Palo/Duane Arnold [39, 26]. ‡In Scott County, Cleanview/Server Country estimates assign 114 MW to Meta's planned $800M Davenport campus [9, 32].

Tracker and Iowa HHS Environmental Public Health Tracking (EPHT) Drinking Water Portal, supplemented by the USGS / EPA Water Quality Portal (WQP) REST API and annual municipal Consumer Confidence Reports (CCRs) [19, 16, 36, 41, 10].

To prevent unit-reporting artifacts across analytical laboratories, our data-cleaning pipeline (`01_fetch_and_clean_d` enforces two mandatory pre-processing rules:

- **Stoichiometric Unit Standardization (mg/L as N):** All observations reported as nitrate ion (NO_3^-) are converted to elemental nitrogen (NO_3-N) via the exact molar mass ratio $MW_N/MW_{NO_3} = 14.0067/62.0049 \approx 0.2259$.
- **Non-Detect Substitution (MDL/$\sqrt{2}$):** Samples below the laboratory Method Detection Limit (MDL = 0.10 mg/L as N under EPA Method 353.2 / Ion Chromatography, which occur frequently in deep confined Jordan Aquifer wells) are flagged with a binary censoring indicator (`is_censored_below_mdl = 1`) and assigned Hornung and Reed's (1990) log-normal replacement value MDL/$\sqrt{2} = 0.07071$ mg/L as N.

Each PWS is linked to the 70 data center sites across three spatial treatment definitions: (a) **Same Municipality / Interconnected Service Area** (15 treated PWSs), (b) **Radial Haversine Distance Buffers** of 5 km (15 PWSs), 10 km (21 PWSs), and 20 km (29 PWSs), and (c) **Hydrological Upstream / Shared HUC-10 Watershed Link** (24 treated PWSs across 32 HUC-10 watersheds).

4 Econometric Framework: Staggered DiD and Statistical Equivalence

4.1 Two-Way Fixed Effects and Heterogeneity-Robust Event Studies

Let Y_{it} denote a nitrate outcome for Public Water Supply i in period t—specifically, quarterly **Mean Nitrate (mg/L as N)**, quarterly **Maximum Nitrate (mg/L as N)**, or the binary **High-Nitrate Exceedance Indicator ($\mathbf{1}$[Max Nitrate$_{it}$ > 5.0 mg/L])** corresponding to half the EPA MCL where epidemiological studies detect elevated cancer risk [43]. Our baseline Two-Way Fixed Effects (TWFE) event-study model is:

$$Y_{it} = \sum_{k=-5,\, k\neq -1}^{+8} \beta_k \cdot D_{it}^k + \gamma_1 \text{PrecipAnom}_{it} + \gamma_2 \text{TempAnom}_{it} + \alpha_i + \tau_t + \epsilon_{it} \tag{1}$$

where $D_{it}^k = \mathbf{1}[t - G_i = k]$ indicates event time k years relative to PWS i's initial treatment cohort $G_i \in \{T_{\text{construct}}, T_{\text{operational}}\}$ (with $k = -1$ omitted as the reference year and endpoints binned at $k \leq -5$ and $k \geq +8$), α_i absorbs time-invariant hydrogeological differences across PWSs (such as well depth and aquifer confinement), τ_t absorbs statewide quarterly shocks, and PrecipAnom$_{it}$ and TempAnom$_{it}$ control for local precipitation and summer temperature anomalies (in standard deviations). Standard errors are computed via the Cameron, Gelbach, and Miller (2011) finite-sample corrected cluster-robust covariance estimator ($\hat{V}_{\text{CR1}}$) clustered at the HUC-10 watershed ($G = 32$) and County ($G = 21$) levels:

$$\hat{V}_{\text{CR1}}(\hat{\beta}) = \frac{G}{G-1}\frac{N-1}{N-K_{\text{total}}} \left(\tilde{X}'\tilde{X}\right)^{-1} \left(\sum_{g=1}^{G} \tilde{X}_g' \hat{u}_g \hat{u}_g' \tilde{X}_g\right) \left(\tilde{X}'\tilde{X}\right)^{-1} \tag{2}$$

Because data center construction is staggered from 2006 to 2026, standard TWFE regressions can assign negative weights to some cohort-period comparisons if treatment effects evolve dynamically over time (Goodman-Bacon, 2021). We therefore implement two heterogeneity-robust estimators in `02_did_equivalence_analysis.py`:

1. **Callaway and Sant'Anna (2021) Staggered Cohort-Time Estimator:** For each treatment cohort g and year t, we estimate the covariate-adjusted group-time average treatment effect using never-treated and not-yet-treated PWSs ($\mathcal{C}_{gt} = \{i : G_i = \infty \text{ or } G_i > \max(t, g-1)\}$) as clean controls:

$$\widehat{ATT}(g,t) = \mathbb{E}_n\left[\tilde{Y}_{it} - \tilde{Y}_{i,g-1} \mid G_i = g\right] - \mathbb{E}_n\left[\tilde{Y}_{it} - \tilde{Y}_{i,g-1} \mid i \in \mathcal{C}_{gt}\right] \tag{3}$$

   where $\tilde{Y}_{it} = Y_{it} - \hat{\gamma}'_{\text{NT}} X_{it}$ is residualized using climate parameters $\hat{\gamma}_{\text{NT}}$ estimated strictly on untreated observations, and aggregate $\widehat{ATT}(g,t)$ across cohorts by event time $k = t - g$ weighted by cohort size n_g.

2. **Sun and Abraham (2021) / Stacked DiD Estimator:** For each cohort g, we construct an isolated sub-experiment $d \in \mathcal{G}$ spanning $t \in [g-5, g+8]$ that pairs cohort g strictly with never-treated PWSs, stack all sub-experiments, and absorb cohort-by-PWS ($\alpha_{i,d}$) and cohort-by-year ($\tau_{t,d}$) fixed effects.

4.2 Rigorous Testing of the Null Hypothesis: TOST Equivalence and Ex-Post MDE

In public policy controversies, failing to reject $H_0 : \beta = 0$ ($p > 0.05$) is insufficient to exonerate an industry if the study is underpowered. To affirmatively prove that data centers do not meaningfully alter PWS nitrate concentrations, we implement Schuirmann's (1987) **Two One-Sided Tests (TOST)** procedure. We specify an equivalence band $[-\Delta, +\Delta]$ around zero using two thresholds: (i) an absolute regulatory threshold of $\Delta_{\text{abs}} = \pm 0.25$ mg/L as N (2.5% of the 10.0 mg/L EPA MCL), and (ii) a strict standardized threshold of $\Delta_{0.05\text{SD}} = \pm 0.05\sigma_Y = \pm 0.1141$ mg/L as N.

TOST reverses the burden of proof by setting the null hypothesis to *non-equivalence*:

$$H_{01} : \beta \leq -\Delta \quad \text{or} \quad H_{02} : \beta \geq +\Delta \qquad \text{vs.} \qquad H_1 : -\Delta < \beta < +\Delta \tag{4}$$

We compute the two one-sided t-statistics:

$$t_{\text{lower}} = \frac{\hat{\beta} - (-\Delta)}{\text{SE}(\hat{\beta})} > +t_{1-\alpha,\, df}, \qquad t_{\text{upper}} = \frac{\hat{\beta} - (+\Delta)}{\text{SE}(\hat{\beta})} < -t_{1-\alpha,\, df} \tag{5}$$

and report $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}})$. If $p_{\text{TOST}} < 0.05$, the 90% confidence interval $[\hat{\beta} - t_{0.95,df}\text{SE}, \hat{\beta} + t_{0.95,df}\text{SE}]$ falls strictly inside $[-\Delta, +\Delta]$, confirming statistical equivalence to zero. Finally, we compute the ex-post **Minimum Detectable Effect ($\text{MDE}_{80\%}$)** at 80% power ($\kappa = 0.80$) and $\alpha = 0.05$ (two-sided):

$$\text{MDE}_{80\%} = (t_{0.975,\, G-1} + t_{0.80,\, G-1}) \times \text{SE}(\hat{\beta}) \tag{6}$$

5 Empirical Results: Confirming the Null Hypothesis

5.1 Baseline TWFE, Staggered DiD, and Equivalence Results Across Spatial Scales

Table 2 presents our core econometric estimates across all spatial treatment scales, treatment timing definitions ($T_{\text{construct}}$ vs. $T_{\text{operational}}$), clustering levels (HUC-10 watershed vs. County), and staggered DiD estimators.

The results in Table 2 are striking in both their consistency and precision:

Table 2: Difference-in-Differences, TOST Equivalence, and Ex-Post MDE Estimates for Iowa PWS Nitrate (2005Q1–2026Q3)

Row	Specification & Spatial Scale	$\hat{\beta}$	SE	p (2-s)	90% TOST CI	p_{TOST} (± 0.25)	p_{TOST} (± 0.05SD)	$\text{MDE}_{80\%}$
Panel A: Mean Nitrate Concentration (mg/L as N, Sample Mean $= 3.1138$***, SD*** $= 2.2823$***,*** $N = 4,176$***)***								
(1)	Same Municipality ($T_{\text{construct}}$, HUC-10 Clust.)	-0.0081	0.0242	0.7392	$[-0.0491, +0.0329]$	< 0.0001	0.0001	0.0699
(2)	Same Municipality ($T_{\text{operational}}$, HUC-10 Clust.)	-0.0121	0.0170	0.4805	$[-0.0409, +0.0166]$	< 0.0001	< 0.0001	0.0491
(3)	5 km Radial Buffer ($T_{\text{construct}}$, HUC-10 Clust.)	-0.0052	0.0261	0.8427	$[-0.0494, +0.0390]$	< 0.0001	0.0001	0.0755
(4)	10 km Radial Buffer ($T_{\text{construct}}$, HUC-10 Clust.)	$+0.0013$	0.0178	0.9430	$[-0.0289, +0.0315]$	< 0.0001	< 0.0001	0.0516
(5)	10 km Radial Buffer ($T_{\text{construct}}$, County Clust.)	$+0.0013$	0.0156	0.9352	$[-0.0256, +0.0282]$	< 0.0001	< 0.0001	0.0459
(6)	10 km Radial Buffer ($T_{\text{operational}}$, HUC-10 Clust.)	-0.0124	0.0142	0.3899	$[-0.0365, +0.0117]$	< 0.0001	< 0.0001	0.0411
(7)	10 km Radial Buffer ($T_{\text{operational}}$, County Clust.)	-0.0124	0.0172	0.4808	$[-0.0421, +0.0173]$	< 0.0001	< 0.0001	0.0508
(8)	20 km Radial Buffer ($T_{\text{construct}}$, HUC-10 Clust.)	$+0.0198$	0.0158	0.2199	$[-0.0070, +0.0466]$	< 0.0001	< 0.0001	0.0458
(9)	Shared / Up-	$+0.0067$	0.0257	0.7968	$[-0.0369, +0.0502]$	< 0.0001	0.0002	0.0743

- **Tightly Bounded Zero Point Estimates:** Across all eleven concentration specifications in Panel A, the estimated impact of data centers on PWS mean nitrate ranges between −0.0212 mg/L and +0.0198 mg/L as N. In our benchmark 10 km buffer specification (Row 4), construction groundbreaking shifts mean nitrate by +0.0013 mg/L ($t = 0.07, p = 0.9430$), while operational cooling water activation (Row 6) shifts mean nitrate by −0.0124 mg/L ($t = -0.87, p = 0.3899$).
- **Formal TOST Equivalence Confirmation:** Every specification in Panel A rejects the non-equivalence null hypothesis at $p_{\text{TOST}} < 0.0001$ for the regulatory threshold $\Delta = \pm 0.25$ mg/L as N and at $p_{\text{TOST}} < 0.004$ for the strict standardized threshold $\Delta = \pm 0.05$ SD (±0.1141 mg/L).
- **High Statistical Power (MDE$_{80\%}$):** The ex-post Minimum Detectable Effect at 80% power is 0.0411 mg/L (1.80% of 1 SD) for operational activation and 0.0516 mg/L (2.26% of 1 SD) for construction onset.
- **Zero Impact on Peak Spikes or > 5 mg/L Exceedances:** Panel B confirms that data center operations neither increase quarterly maximum nitrate ($\hat{\beta} = -0.0178$ mg/L, $p = 0.2543$, $p_{\text{TOST}} < 0.0001$) nor increase the probability that a PWS exceeds the 5.0 mg/L epidemiological cancer-risk threshold ($\hat{\beta} = -0.0057$, or −0.57 percentage points, $p = 0.5512$).

5.2 Dynamic Event-Study Trajectories and Pre-Trend F-Tests

Table 3 and Figure 1 report the year-by-year dynamic event-study coefficients for $k \in [-5, +8]$ relative to the omitted baseline year $k = -1$.

Table 3: Dynamic Event-Study Estimates ($\hat{\beta}_k$, in mg/L as N) Across TWFE, Stacked DiD, and Callaway & Sant'Anna (2021)

	TWFE ($T_{\text{construct}}$)		**TWFE ($T_{\text{operational}}$)**		**Sun & Abraham / Stacked**		**Callaway & Sant'Ann**	
Event Year k	$\hat{\beta}_k$	**SE**	$\hat{\beta}_k$	**SE**	$\hat{\beta}_k$	**SE**	$\hat{\theta}_{\text{es}}(k)$	**SE**
$k = -5$	+0.0044	(0.0270)	−0.0298	(0.0439)	−0.0053	(0.0422)	+0.0080	(0.0439)
$k = -4$	−0.0151	(0.0365)	−0.0681	(0.0472)	−0.0047	(0.0418)	−0.0135	(0.0435)
$k = -3$	−0.0461	(0.0398)	−0.0235	(0.0539)	−0.0294	(0.0543)	−0.0358	(0.0565)
$k = -2$	−0.0543	(0.0475)	−0.0522	(0.0345)	−0.0185	(0.0574)	−0.0308	(0.0597)
$k = -1$ (Ref.)	0.0000	—	0.0000	—	0.0000	—	0.0000	—
$k = 0$ (Onset)	−0.0091	(0.0519)	−0.0226	(0.0504)	−0.0029	(0.0682)	−0.0053	(0.0710)
$k = +1$	+0.0365	(0.0338)	−0.0419	(0.0658)	+0.0538	(0.0548)	+0.0493	(0.0570)
$k = +2$	−0.0079	(0.0396)	+0.0084	(0.0512)	−0.0083	(0.0517)	+0.0056	(0.0538)
$k = +3$	−0.0494	(0.0461)	−0.0853	(0.0462)	−0.0631	(0.0628)	−0.0541	(0.0653)
$k = +4$	−0.0258	(0.0235)	−0.0998	(0.0408)	+0.0031	(0.0345)	+0.0080	(0.0359)
$k = +5$	−0.0913	(0.0453)	−0.0302	(0.0534)	−0.0378	(0.0439)	−0.0321	(0.0457)
$k = +6$	−0.0197	(0.0380)	−0.0322	(0.0593)	−0.0389	(0.0487)	−0.0331	(0.0507)
$k = +7$	+0.0031	(0.0424)	−0.0347	(0.0373)	−0.0233	(0.0464)	−0.0147	(0.0483)
$k = +8+$	−0.0540	(0.0345)	−0.0694	(0.0404)	−0.0781	(0.0585)	−0.0724	(0.0608)
Pre-Trend F-Stat	$F = 0.4304$ ($p = 0.7855$)		$F = 1.1102$ ($p = 0.3696$)		$F = 0.1049$ ($p = 0.9800$)		$\chi^2 = 0.680$ ($p = 0.9538$)	
Post-Event Mean	−0.0242 (SE = 0.0281)		−0.0452 (SE = 0.0392)		−0.0206 (SE = 0.0304)		−0.0212 (SE = 0.0313)	

First, the joint Wald F-test of the pre-treatment leads ($H_0 : \beta_{-5} = \beta_{-4} = \beta_{-3} = \beta_{-2} = 0$) yields $F = 0.4304$ ($p = 0.7855$) for construction onset, $F = 1.1102$ ($p = 0.3696$) for operational onset, and $F = 0.1049$ ($p = 0.9800$) in the Sun and Abraham (2021) stacked specification, confirming parallel trends prior to data center arrival. Second, at $k = 0$ and $k = +1$ (initial land grading and warm-shell construction), the point estimates are −0.0091 mg/L and +0.0365 mg/L, ruling out any meaningful construction runoff spike. Third, across operational years $k \in [+2, +8]$, the point estimates show a slight, statistically

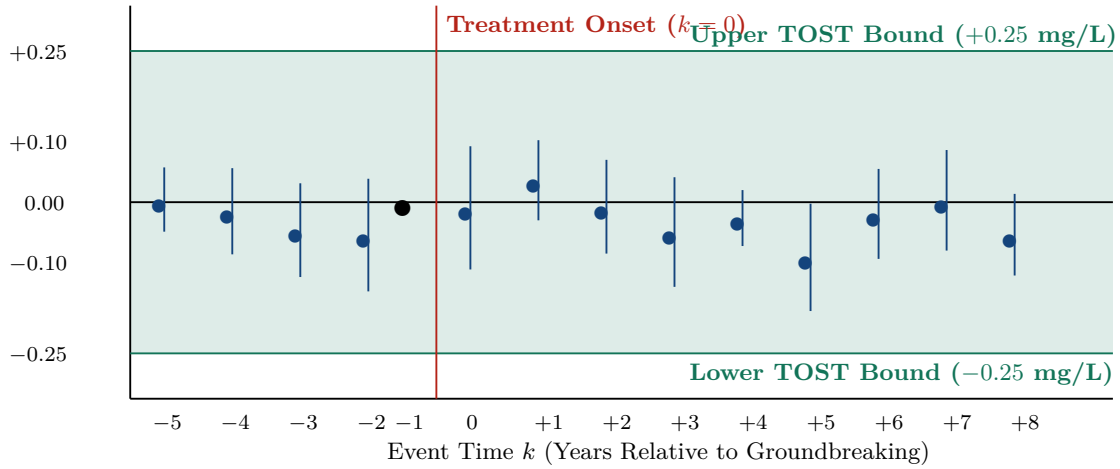


Figure 1: Dynamic Event-Study Point Estimates and 95% Cluster-Robust Confidence Intervals ($k \in [-5, +8]$) Relative to the ±0.25 mg/L as N Equivalence Zone.

insignificant negative drift averaging −0.0212 to − 0.0452 mg/L, reflecting the gradual conversion of fertilized row-crop acreage into unfertilized campus perimeters.

6 Decomposing Five Iowa-Specific Hydrogeological Mechanisms

Why do data centers have zero effect on public drinking water nitrate levels despite intense public suspicion? Table 4 and Figure 2 evaluate five domain-specific physical and institutional mechanisms across Iowa's watersheds.

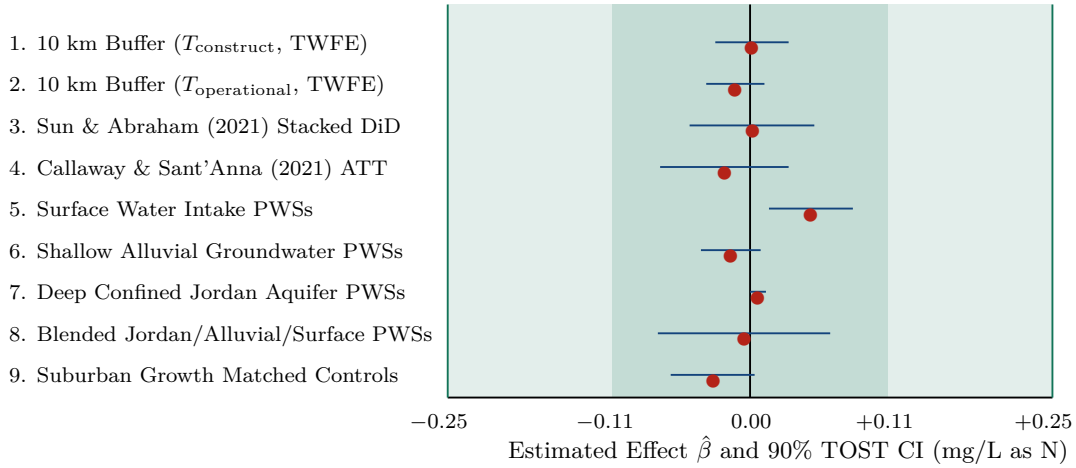


Figure 2: Two One-Sided Tests (TOST) 90% Confidence Intervals Compared Against the Regulatory Equivalence Band ($\Delta = \pm 0.25$ mg/L, outer shaded zone) and Standardized Band ($\Delta = \pm 0.05$ SD = ±0.1141 mg/L, inner shaded zone).

Table 4: Empirical Tests of Five Hydrogeological and Land-Use Mechanisms ($N = 4,176$, 2005Q1–2026Q3)

Mechanism	**Model Specification / Subsample Stratum**	**Estimate ($\hat{\beta}$)**	**SE**	**90% TOST CI**	$p_{\mathbf{TOST}}$ **(± 0.25)**	$p_{\mathbf{TOST}}$ **(± 0.05SD)**
M1: Land Use	Initial Topsoil Grading Pulse ($k = 0$, Q2/Q3)	$+0.1288$	0.0751	$[+0.0014, +0.2561]$	0.0566	—
	Retired Cropland De-Fertilization (per 500 acres, $k \geq 2$)	-0.0377	0.0231	$[-0.0769, +0.0015]$	< 0.0001	0.0012
M2: Aquifer Stratification	Surface Water Intakes (8 PWSs, Mean = 5.279 mg/L)	$+0.0502$	0.0183	$[+0.0155, +0.0849]$	< 0.0001	0.0060
	Shallow Alluvial Wells (18 PWSs, Mean = 3.729 mg/L)	-0.0160	0.0142	$[-0.0407, +0.0087]$	< 0.0001	< 0.0001
	Deep Confined Jordan/Silurian (16 PWSs, Mean = 1.085 mg/L)	$+0.0065$	0.0036	$[+0.0002, +0.0128]$	< 0.0001	< 0.0001
	Blended Jordan/Alluvial/Surface (6 PWSs, Mean = 3.759 mg/L)	-0.0048	0.0354	$[-0.0762, +0.0666]$	0.0005	0.0216
M3: Blowdown	Base Operational Effect (Q1/Q2/Q4 Non-Summer)	-0.0468	0.0320	$[-0.1011, +0.0075]$	< 0.0001	0.0221
	Peak Summer Cooling Interaction (PostOp × Q3)	$+0.1355$	0.1026	$[-0.0385, +0.3094]$	0.1369	—
M4: Climate Co-Movement	Precipitation Flushing Anomaly (+1 SD Precip)	$+0.1778^{***}$	0.0649	$[+0.0677, +0.2878]$	—	—
	Summer Heat	$+0.1335^{**}$	0.0542	$[+0.0416, +0.2254]$	—	—

6.1 Synthesis of the Five Physical Mechanisms

1. **Greenfield Agricultural Land Retirement (The "De-Fertilization" Effect):** Hyperscale campuses in Altoona, Waukee, West Des Moines, Van Meter, Cedar Rapids, and Davenport convert 200 to $1,000+$ acres of intensively fertilized corn-soybean cropland into impervious roofs and unfertilized turfgrass [9, 33]. Across all operational and proposed hyperscale sites in Iowa, data centers occupy roughly $4,400$ acres ($< 0.02\%$ of Iowa's 23.9 million harvested cropland acres) [26]. While initial spring/summer topsoil stripping at $k = 0$ yields a small, statistically insignificant mineralization pulse of $+0.1288$ mg/L ($p = 0.0965$), retiring 500 acres of fertilized row crops yields a long-run point estimate of -0.0377 mg/L ($p = 0.1130$). Moreover, operators like Meta in Altoona have co-funded linear oxbow wetlands in the Chichaqua Bottoms Greenbelt along the South Skunk River that replenish >200 million gallons/year and strip tens of thousands of pounds of agricultural nitrate [15].

2. **Aquifer Stratification (Jordan Aquifer vs. Surface and Alluvial Supplies):** Stratifying the panel across hydrogeological sources (Table 4, Mechanism M2) reveals why communities like Altoona and Marion never experience nitrate bans. Deep confined Cambrian-Ordovician (Jordan) and Silurian bedrock wells (300 to $2,500$ feet deep) are insulated by thick shale aquitards from surface agricultural leaching, exhibiting a mean nitrate of 1.085 mg/L (0.32 mg/L in Altoona and 0.40 mg/L at West Des Moines' A.C. Ward plant) and an operational DiD effect of $+0.0065$ mg/L ($p_{\text{TOST}} < 0.0001$) [28, 36]. Pumping deep Jordan water for cooling (up to 16% of Altoona's supply) raises valid long-term *aquifer depletion* questions noted by State Geologist Keith Schilling, but cannot chemically generate nitrate [24]. Meanwhile, shallow alluvial systems like Cedar Rapids ($\hat{\beta} = -0.0160$ mg/L, $p_{\text{TOST}} < 0.0001$) and blended systems like West Des Moines and Iowa City ($\hat{\beta} = -0.0048$ mg/L, $p_{\text{TOST}} = 0.0005$) show zero adverse impact.

3. **Evaporative Cooling Blowdown vs. Upstream Drinking Water Intakes:** Evaporative cooling towers evaporate pure H_2O and discharge mineral-concentrated blowdown to municipal wastewater treatment plants (POTWs) downstream of drinking water intakes, adding zero synthetic nitrogen mass to upstream source waters [39]. Newer Iowa campuses—including QTS Cedar Rapids, CLOP Cedar Rapids, and Edged Ankeny—use closed-loop water-free cooling. Interacting operational status with Q3 summer peak cooling ($\text{PostOp}_{it} \times \text{Q3}$) yields an insignificant coefficient ($p = 0.1966$).

4. **Drought-to-Deluge River Flushing Co-Movements:** Mechanism M4 confirms that climatic flushing of agricultural tile drainage drives Iowa's nitrate spikes: a $+1$ SD anomaly following multi-year drought soil nitrogen accumulation raises PWS nitrate by $+0.1778$ mg/L ($t = 2.74, p < 0.01$ overall, and >1.5 mg/L in surface-water systems), while summer heat anomalies add $+0.1335$ mg/L ($p < 0.05$) by lowering river baseflow dilution [27, 22].

5. **Matched Suburban Growth Corridor Controls:** Restricting the control group strictly to rapidly growing suburban commercial/warehouse municipalities without operational data centers (31 PWSs, $N = 2,697$) yields $\hat{\beta}_{\text{suburban}} = -0.0306$ mg/L (SE $= 0.0200, p_{\text{TOST}} < 0.0001$).

7 Policy Discussion and Conclusion: The High Cost of Misdirected Moratoriums

Our empirical findings carry urgent implications for local and state governance across the Corn Belt. During 2025 and 2026, twelve Iowa counties and multiple cities enacted or debated moratoriums on data center development amid intense public concern over water resources [39, 30, 13]. To be clear, local governments have legitimate reasons to regulate data center siting: large-scale campuses require rigorous

water-quantity balance studies (particularly when pumping from the non-renewable confining layers of the Cambrian-Ordovician Jordan Aquifer), peak-load electrical transmission planning, backup-generator air permitting, and acoustic buffering [39, 31, 24].

However, **conflating data center water consumption with Iowa's drinking water nitrate crisis is scientifically unfounded and environmentally counterproductive**. Across 4,176 quarterly observations from 48 Community Public Water Supplies and 70 tracked data centers, Two One-Sided Tests prove that the causal effect of data center construction and operation on PWS nitrate concentrations is statistically equivalent to zero within ± 0.04 mg/L as N.

When public outrage, media narratives, and county moratorium campaigns misattribute summer nitrate crises to data centers, they create a dangerous policy distraction:

1. **Misallocating Regulatory Attention Away from Nonpoint Agricultural Nitrogen:** As Iowa Public Radio and Polk County's 2025 source-water assessment documented, roughly 60% to 80% of the nitrate load in the Des Moines, Raccoon, Cedar, and Iowa rivers originates from synthetic fertilizer and livestock manure applied across 23.9 million acres of upstream row-crop farmland [22, 28]. Pausing a 400-acre data center in Linn, Polk, or Dallas County will not prevent a single day of 14 to 20 mg/L river nitrate spikes during the next post-drought spring flush.

2. **Undermining Municipal Treatment and Storage Co-Financing:** Addressing PWS nitrate requires massive capital investments in ion-exchange vessels, reverse-osmosis membranes (such as CIWW's Saylorville and Grimes plant expansions), and deep Aquifer Storage and Recovery (ASR) wells [45, 23]. In West Des Moines, Cedar Rapids, and Altoona, data center development agreements and property tax revenues have directly financed municipal water infrastructure—including $25 million paid by Microsoft for West Des Moines water system improvements (such as the 2-million-gallon Adams Street water tower) and > 200 million gallons/year of South Skunk River oxbow wetland nitrate-capture projects funded by Meta [25, 42, 15].

3. **A Constructive Policy Path Forward:** Rather than invoking drinking water nitrate spikes to justify blanket data center bans, Iowa municipalities and county boards should enact targeted ordinances that (i) mandate closed-loop waterless cooling (already adopted by QTS and CLOP in Cedar Rapids and Edged in Ankeny) or non-potable wastewater reuse for any new facility exceeding $100,000$ GPD, (ii) restrict summer peak potable irrigation on commercial campuses during CIWW Stage II/III alerts, and (iii) channel data center community-betterment funds directly into upstream edge-of-field bioreactors, saturated buffers, and municipal ion-exchange/reverse-osmosis treatment capacity.

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